

Tao Su (苏桃)

✉ sutao@bimsa.cn

<https://tao-su-math.github.io/>



Employment

- Jan.2024-present **Assistant professor**, *BIMSA*, Beijing
- Sep.2022-Sep.2023 **Postdoc**, *IMS, CUHK*, Hong Kong
- Jan.2021-Jul.2022 **Postdoc**, *YMSC*, Beijing
supported by the International Postdoctoral Exchange Fellowship (Talent-Introduction)
- Sep.2018-Aug.2020 **Postdoc**, *École Normale Supérieure*, Paris
supported by the ANR Microlocal Project

Education

- 2018 **Ph.D. in Mathematics**, *UC Berkeley*
Advisors: Vivek Shende, and Richard E. Borcherds
- 2012 **B.S. in Mathematics and Physics**, *Tsinghua University*
Fellowship in the Tsinghua Xuetao Talents Program

Research interests

- Algebraic analysis, Algebraic Geometry, and Symplectic Geometry.
- Topics of interest include: Legendrian knots and constructible sheaves, Hodge theory of character varieties and various $P=W$ conjectures, Zariski cancellation problem in characteristic 0, Planar curve singularities and link homology, Coulomb branch algebras and affine Springer theory, microlocal sheaves and mirror symmetry.

Publications/Preprints

- Log-concavity from enumerative geometry of planar curve singularities, with Baiting Xie, Chenglong Yu, arXiv:2603.27888. (*submitted*)
- Purity of generalized affine Springer fibers from generic planar curve singularities, with Taiwang Deng, arXiv:2509.20800. (*under revision*)
- Integral cohomology of dual boundary complexes is motivic, arXiv:2408.17301.
- Cell decomposition and dual boundary complexes of character varieties, **Advances in Mathematics**, 498 (2026), 111043
- Dual boundary complexes of Betti moduli spaces over the two-sphere with one irregular singularity, **Advances in Mathematics**, 462 (2025), 110101.
- Augmentations are sheaves for Legendrian graphs, with Byung Hee An, Youngjin Bae, **Journal of Symplectic Geometry**, 20 (2022), pp.259-416.
- Augmentations and ruling polynomials for Legendrian graphs, with Byung Hee An, Youngjin Bae, **Algebraic & Geometric Topology**, 22 (2022), pp.2079-2185.

- A Hodge-theoretic study of augmentation varieties associated to Legendrian knots/tangles, **Ph.D thesis**, ProQuest 2018.
- Ruling polynomials and augmentations for Legendrian tangles, arXiv:1707.04948.

Invited talks

- Jun.8-11,2026 4 lectures on cell decomposition and dual boundary complexes of character varieties, Institute for Theoretical Sciences, Westlake University, Hangzhou.
- May.14,2026 Number Theory Lunch Seminar, BIMS, Beijing.
- May.7,2026 Number Theory Lunch Seminar, BIMS, Beijing.
- Feb.5,2026 MIST Conference, IMS, CUHK, Hong Kong.
- Jan.6,2026 Invited lecture (45 minutes), the 10-th International Congress of Chinese Mathematicians, SIMS, Shanghai.
- Sep.22,2025 Conference on Geometry and Mathematical Physics, Nanjing University of Science and Technology.
- Sep.19,2024 Conference on Topological Strings on Calabi-Yau Manifolds - Math and Physics, BICMR, Beijing.
- Jun.28,2024 Geometry and Topology seminar, Institute of Geometry and Physics, USTC, Hefei.
- May 11,2024 Symplectic geometry and Contact topology Conference, Tianjin University.
- Feb.22,2024 BIMS AG seminar, BIMS, Beijing.
- Nov.13,2023 CAS seminar, AMSS, CAS, Beijing.
- Oct.8-Oct.16,2023 5 lectures on dual boundary complexes of character varieties, YMSC, Beijing.
- Aug.2,2023 MIST workshop, IMS, CUHK, Hong Kong.
- Jan.7,2023 MIST workshop, IMS, CUHK, Hong Kong.
- May 25,2022 Zoom seminar, Zhejiang University.
- Apr.21,2022 Algebraic Geometry/Number theory seminar (zoom), Boston College.
- Dec.17,2021 Tsinghua Algebraic Geometry workshop, YMSC, Beijing.
- Dec.14,2021 Symplectic Geometry and Mathematical Physics seminar, BICMR, Beijing.
- Nov.29,2021 GeoTop seminar, YMSC, Beijing.
- Nov.10,2021 Geometry and Topology seminar, AMSS, Chinese Academy of Sciences, Beijing.
- Oct.13,2021 Geometry and Topology seminar, Chern Institute of Mathematics, Tianjin.
- Oct.12,2021 Geometry and Topology seminar, Chern Institute of Mathematics, Tianjin.
- Sep.24,2021 Geometric representation seminar, YMSC, Beijing.
- Mar.24,2021 Moscow-Beijing topology seminar, Zoom talk.
- Jul.31,2019 Director's seminar, IBS-CGP, South Korea.
- Jun.27,2019 Conference on Homological algebra, Microlocal analysis, and Symplectic geometry, CRM, Montreal.
- Jan.19,2019 Conference on Number theory, Algebra, Combinatorics, IAS of HIT, Haerbin, China.
- Nov.30,2018 Symplectic seminar, RIMS, Kyoto.
- Nov.9,2018 Symplectic seminar, ENS, Paris.
- Oct.5,2018 Topology seminar, Fourier Institute, Grenoble.
- May 24,2018 Student geometry/topology seminar, UC Davis.

Mar.6,2018 EGN postdoc and student seminar, MSRI.
 Dec.5,2017 3-manifold Seminar, UC Berkeley.
 Jun.28,2017 AIM square meeting “Sheaf theory and Legendrian knots”, San Jose.

Services

Organized Geometry, Representations, And Some Physics (GRASP) seminar, BIMS, Spring 2024-present.
 Organized Derived Algebraic Geometry learning seminar, YMSC, Spring 2021.
 Tutored Friend scholar for Qiuzhen college, Tsinghua university, Apr. 2021-Jun.2022.
 Referee for Symmetry, Integrality and Geometry: Methods and Applications.

Academic visits

Jun.7-13, 2026 Institute for Theoretical Sciences, Westlake University, Hangzhou, China
 Jun.27, 2024-Jul.5, 2024 Institute of Geometry and Physics, USTC, Hefei, China
 Oct.30, 2023-Jan.30, 2024 Academy of Mathematics and Systems Science, CAS, China
 Oct.7- Oct.17,2023 YMSC, Beijing, China
 Sep.01- Nov.25,2020 IHES, Paris, France
 Jul.28- Aug.04,2019 IBS-CGP, South Korea
 Jun.08- Jun.30,2019 CRM, Montreal, Canada
 Nov.25- Dec.09,2018 RIMS, Kyoto, Japan
 Oct.04- Oct.06,2018 Fourier Institute, Grenoble, France

Honors and Awards

2025 Ruolin paper award, BIMS.
 2021 International Postdoctoral Exchange Fellowship, China Postdoc Science Foundation.
 2017 Graduate Division Summer Grant, UC Berkeley.
 2015 Graduate Division Summer Grant, UC Berkeley.
 2009-2012 Fellowship in the Tsinghua Xuetao Talents Program, Tsinghua University.
 2011 S.-T. Yau College Students Mathematics Contests:
 ○ (All-around Competition) Bronze medal ○ (Geometry and Topology) Bronze medal
 ○ (Algebra, Number Theory, and Combinatorics) Bronze medal ○ (Group competition) Silver medal

Teaching experience

- Fall 2026 Lectures on geometric representation theory 0 (linear algebraic groups) - part 2, BIMS.
- Spring 2026 Lectures on geometric representation theory 0 (linear algebraic groups) - part 1, BIMS.
- Fall 2025 Lectures on hypersurface singularities III, BIMS.
- Spring 2025 Lectures on hypersurface singularities II, BIMS.
- Fall 2024 Lectures on hypersurface singularities I, BIMS.
- Spring 2018 TA for Math 113, Introduction to Abstract Algebra, UC Berkeley.
- Fall 2017 GSI for Math 16B, Analytic Geometry and Calculus, UC Berkeley.
- Fall 2016 GSI for Math 16B, Analytic Geometry and Calculus, UC Berkeley.
- Spring 2016 GSI for Math 16B, Analytic Geometry and Calculus, UC Berkeley.
- Fall 2015 GSI for Math 16B, Analytic Geometry and Calculus, UC Berkeley.
- Spring 2015 GSI for Math 54, Linear Algebra and Differential Equations, UC Berkeley.
- Fall 2014 GSI for Math 54, Linear Algebra and Differential Equations, UC Berkeley.
- Spring 2014 GSI for Math 16B, Analytic Geometry and Calculus, UC Berkeley.
- Fall 2013 GSI for Math 1B, Calculus, UC Berkeley.